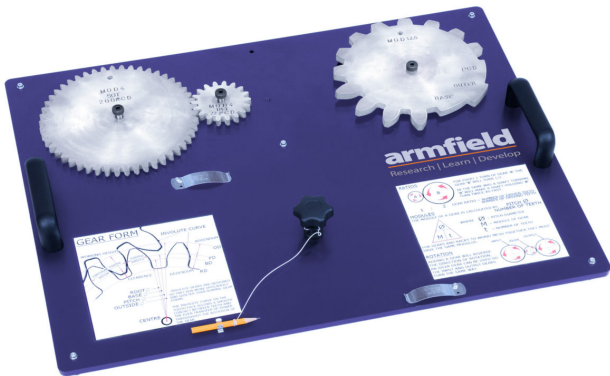




SD-1.53 – Gear Tooth Form Apparatus

Experimental determination of shape and construction of an involute curve

PRODUCT CODE: **SD-1.53**

Categories: Educational, SD-1.53-Gear Tooth Form Apparatus, SV - Theory Of Machines, SV Series - Statics & Vibrations

Tags: Gear Geometry , Gear Tooth Form Apparatus , Involute Gear Form And Creation , Meshing , Module , Number Of Teeth , Pitch Circle Diameter (PCD) , Ratios , SD-1.53 , Spur Gears

DESCRIPTION

A bench top base plate contains all elements for this experiment.

Three gears are located onto the base plate. A large gear and small gear mesh together and can be rotated to produce analysis of ratios. A third gear is removable to allow students to review its tooth form, pitch circle diameter (PCD) and other key geometry of a tooth form.

The apparatus also explains the form of an involute curve and how this is used to create a gear tooth profile. To help describe what an involute curve is, paper can be placed into the apparatus and an involute curve drawn. A set of vernier callipers is provided to allow the removable gear to be measured to help give a practical analysis of the gear form. Labels are used on the unit to help clearly explain gear theory.

TECHNICAL SPECIFICATIONS

- Large (Mod 4) gears
- **Large gear:** 50 tooth, Module 4, 200mm PCD
- **Small gear:** 18 tooth, Module 4, 72mm PCD
- Removable gear module 12.5

FEATURES & BENEFITS

- Bench or wall mounted apparatus
- Two connecting gears supplied
- One gear for measuring
- Drawing of Involute Gear form
- Theory and terminology clearly labelled on unit

Experimental Content

- Experimental determination of shape and construction of an involute curve
- Explanation of Gear Form
- Explanation of Gear Ratios
- Explanation of Gear Modules
- Explanation of Gear Trains and Rotational Direction

Ordering Specification

- 1 x **SD-1.53** assembly
- 1 x Digital vernier
- Spare cord
- 5 x Pencil
- Instruction manual
- Packing list
- Test sheet

Series Includes:

Theory of Machines

- SD-1.50 Wheel and Axle
- SD-1.51 Wheel and Differential Axle Apparatus
- SD-1.52 Efficiency of Screw Threads
- SD-1.53 Gear Tooth Form Apparatus
- SD-1.54 Cam and Follower Apparatus
- SD-1.55 Simple Flywheel
- SD-1.56 Epicyclic Gear Train
- SD-1.57 Castor, Camber, and King Pin Inclination
- SD-1.58 Relation Between Angular and Linear Speeds
- SD-1.59 Disc Brake Apparatus
- SD-1.60 Static & Dynamic Balancing Apparatus
- SD-1.61 Governor Apparatus
- SD-1.62 Gear Assembly Unit Combined Drives
- SD-1.63 Single Epicyclic Gear Train

Requirements

Sturdy bench top

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.03m³

Gross Weight: 13kg



Dimensions

Length: 0.60m

Width: 0.53m

Height: 0.08m

Order Codes

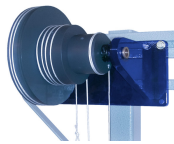
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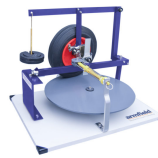
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